Jasmin Støvlbæk

Bioinformatics Engineer (MSc Eng) Cancer Genomics & Multi-omics

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Education

MSc Eng, Bioinformatics and Systems Biology – DTU | 2024–2026
- Thesis: Isoform- and mutation-level contributions to cancer survival through multi-omics integration
- Built analysis-ready TCGA/CPTAC cohorts with endpoints + clinical covariates; delivered reproducible reports

BSc, Life Science Engineering – DTU | 2020–2023
- Thesis: Cancer hallmark analysis in RNA-seq data (TCGA breast cancer; GSVA/ssGSEA)

The Academy for Talented Youth | 2013 - 2015
Talent Program

Summary

Bioinformatics engineer specialising in integrative analysis of large-scale omics datasets to support biomarker discovery and translational research. Experienced in building end-to-end computational pipelines for transcriptomics, proteomics and mutation-derived features, linking molecular data with clinical outcomes through survival modelling and multi-omics integration. Strong focus on reproducible workflows, interpretable results and collaboration across computational, biological and clinical teams.

Research & Experience

Master's Thesis - Isoform analysis & cancer survival (DTU) | 2026
- Developed end-to-end survival modelling workflows (CoxPH + high-dimensional extensions) for gene/isoform features
- Integrated transcriptomics, proteomics and mutation-derived signals with clinical covariates to identify survival-associated biomarker candidates
- Automated QC, feature filtering, and publication-style visualisations to support biological prioritisation
- Maintained codebase with Git; wrote a paper-style thesis manuscript and method documentation

Isoform-resolved database of cancer proteomics | 2025
- Designed a structured, queryable database integrating isoform-level proteomics quantifications and metadata
- Prioritised consistent identifiers, traceability, and downstream analysis usability

Deep Learning Course Project - DTU | 2024
- Built and trained a convolutional neural network for image classification.
- Applied GradCam for explainability, highlighting decision-driving features.

Bachelor's Thesis - Enabling Analysis of Cancer Hallmarks | 2023
- Developed a novel hallmark-associated gene set collection and validated with RNA-seq breast cancer data from TCGA.
- Applied clustering, differential expression, and enrichment analyses to demonstrate biological relevance and cancer heterogeneity.

Exhibition Pilot - Experimentarium | 2019 - 2021
- Delivered engaging science communication and interactive demonstrations, simplifying complex scientific concepts.
- Developed strong public speaking and communication skills.

Sports & Team Leadership
- Former elite football player and trained football coach.
- Strong teamwork, resilience, and leadership experience transferable to interdisciplinary research environments.

Technical skills

Computational & Statistical Analysis
- Survival modelling (Cox proportional hazards)
- Integrative multi-omics analysis
- Biomarker discovery in clinical cohorts
- Isoform / splicing-aware transcriptomics
- Feature engineering and cohort harmonisation

Programming & Data Science
- R (Bioconductor, survival, tidyverse)
- Python (pandas, scikit-learn, matplotlib)
- Bash, Git, Linux

Bioinformatics & Workflows
- RNA-seq analysis and gene expression modelling
- Mutation-derived feature integration
- Reproducible analysis pipelines
- HPC workflows (SLURM)
- Workflow management (Snakemake – basic)

Highlights

End-to-end bioinformatics pipelines for translational datasets
(QC → cohort harmonisation → multi-omics integration → survival modelling → reproducible reporting)

Integrative biomarker prioritisation
Combined mutation-derived evidence with expression context to rank biologically relevant candidates

Isoform-resolved transcriptomics and splicing-aware analyses
Supporting biomarker discovery and target interpretation

Cross-disciplinary collaboration
Comfortable working across computational, biological and clinical teams; strong scientific communication

Clinical cohort analysis
Experience integrating multi-omics features with clinical endpoints in TCGA/CPTAC cancer cohorts

Sequencing-to-insight mindset
Motivated by workflows linking sequencing data, in-silico prioritisation and downstream biological validation